SPE-229172-MS

# Leveraging Aquifer Thermal Storage in Shallow Aquifers with Sustainable Low Temperature Resources

M. Kamali, Petroleum Technology Research Centre, Regina, SK, Canada; A. Rangriz Shokri, University of Alberta, Edmonton, AB, Canada; E. Nickel and Z. Movahedzadeh, Petroleum Technology Research Centre, Regina, SK, Canada; N. Jia and A. Veawab, University of Regina, Regina, SK, Canada; R. Narayanasamy, Petroleum Technology Research Centre, Regina, SK, Canada; R. Chalaturnyk, University of Alberta, Edmonton, AB, Canada



This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.



## Abstract

Residential heating and hot water demand often rely on fossil fuels, increasing $CO_2$ emissions. Geothermal energy has emerged as a renewable alternative, with seasonal thermal storage playing a key role. Aquifer thermal energy storage (ATES) stores excess energy underground during peak supply periods and extracts it when needed. This paper explores ATES's potential to improve efficiency and lower emissions by utilizing solar heat and surplus geothermal energy for sustainable residential heating and hot water supply.

This study utilizes a numerical model of the Mannville Aquifer to simulate energy storage dynamics. Heated water from solar thermal and excess deep geothermal heat is injected into the reservoir to charge it during low-demand periods. The numerical model examines temperature fluctuations in the aquifer during charge and discharge cycles, assessing its effectiveness. Additionally, it explores the feasibility of integrating solar thermal energy into an open-loop shallow aquifer storage system, aiming to improve efficiency and sustainability in geothermal energy utilization through a deeper investigation into solar thermal inclusion.

Numerical results demonstrate that shallow aquifers play a crucial role in regulating heat production from deep geothermal reservoirs. By balancing energy extraction between solar energy sources and shallow and deep geothermal reservoirs during peak demand, the system maintains efficiency while stabilizing storage temperatures. Excess heat stored in shallow aquifers during high supply periods (e.g., abundant solar energy in summer) reduces temperature decline and preserves thermal energy for future use. This approach enhances sustainability by reducing overall energy consumption while optimizing energy utilization. Findings indicate that integrating excess deep geothermal heat storage with solar energy enhances long-term efficiency and system stability. Utilizing shallow aquifer storage allows urban energy solutions to harness solar energy in summer and geothermal energy in winter, reducing reliance on conventional heating systems while refining renewable energy management. The numerical model presents a practical method for integrating such energy distribution, making it more effective for large-scale urban applications. This study underscores the potential of shallow aquifer storage in advancing sustainable energy systems, providing a promising avenue for efficient solar and geothermal energy utilization in densely populated areas.

Results of this study showcase the effectiveness of combining solar and geothermal technologies to optimize renewable energy use and support sustainable urban heating solutions. The insights from the modeling workflow can be used to improve energy storage and extraction methods.

## Introduction

### Growth of Sustainable Heating Technologies

The global challenge of climate change is tightly connected to developing better strategies for reducing greenhouse gas (GHG) emissions, including $CO_2$ (Movahedzadeh et al., 2021). Heating and cooling, which relies heavily on fossil fuels, accounted for approximately 40% of global energy demand during 2015-2019, and significantly contributed to GHG emissions. Transitioning from conventional heating methods to more sustainable energy sources in the heating sector is a critical step toward decarbonization. Renewable and low carbon energy sources, including solar (especially solar thermal), biomass, waste heat, direct-use geothermal, $CO_2$ geothermal, and power-to-heat systems such as heat pumps, have contributed substantially to heat energy supply and are expected to play an increasingly important role in the future (Beernink et al., 2024; Daniilidis et al., 2022; Rangriz Shokri et al., 2021). Today, district heating networks serve residential settings across various regions and can be conveniently retrofitted with sustainable heating and cooling systems (Zwamborn et al., 2022).

### Challenges of Seasonal Demand and Supply

Despite their potential, renewable energy systems face challenges, such as high upfront costs and a mismatch between seasonal energy demand and resource availability, especially for solar energy (Daniilidis et al., 2017, 2022; Zwamborn et al., 2022). Geothermal energy has a long history of utilization, and its associated technical challenges are well-understood. Studies underscore the importance of efficient and strategically managed geothermal production for sustainable development (Daniilidis et al., 2017). Geothermal energy serves as a reliable baseload resource, whereas solar offers intermittent heat supply.

In Canada, urban areas receive approximately 5.3 GJ/$m^2$ of annual global solar radiation. However, a key barrier to solar adoption is its scarcity during peak heating demand seasons. In this context, thermal energy storage (TES) offers an effective solution for synchronizing energy supply and demand in line with seasonal fluctuations, a key element toward sustainable development of renewable energies (Daniilidis et al., 2017, 2022; Mesquita et al., 2017; Sibbitt et al., 2012).

### TES Technologies and Aquifer Thermal Storage

Thermal energy storage (TES) technologies such as borehole thermal energy storage (BTES) and aquifer thermal energy storage (ATES) are gaining attention for their role in decarbonization. BTES operates via closed-system tubing systems, while ATES exchanges heat via convection within aquifers. Like all technologies, ATES has advantages, including large storage capacity, minimal surface footprint, flexible energy management, low operational costs, and promising profitability, as well as challenges. These challenges include the need for suitable aquifers near dense urban centres, finding proper aquifers with adequate subsurface properties, managing temperature limits for heat harvesting and distribution, geochemical risks such as salt precipitation and scaling, and optimizing the ratio of energy production to storage (Daniilidis et al., 2022; Talman et al., 2020). Heat stored in aquifers may originate from surplus geothermal energy, solar energy, or industrial waste heat (Vrijlandt et al., 2023).

A notable example in Canada is the Drake Landing Solar community in Okotoks, Alberta, which utilized BTES with 144 boreholes. Using 2,293 $m^2$ of flat-plate solar collectors during spring, summer, and fall, heat is stored to supply energy for 52 housing units throughout winter. This project demonstrates the reliability of solar heat harvesting and seasonal thermal storage for district heating in a region with severe winters (Mesquita et al., 2017; Sibbitt et al., 2012).

Solar collectors work by absorbing solar heat and transferring it to a flowing medium. In addition to flat-plate solar collectors, evacuated tube solar collectors offer better performance during low radiation periods and cold days. It is reported that the temperature of the collecting medium can reach temperatures in the range of 50-200 °C (Siuta-Olcha et al., 2020).

**Study Focus and Objectives**

This study investigates the utilization of solar thermal energy via evacuated tube collectors in the City of Regina, Saskatchewan, based on the region's solar irradiation data. The collected heat is stored seasonally in an aquifer through an injection well and is extracted during peak demand periods via a production well. Energy demand data from a representative facility is used to evaluate the aquifer's storage and retrieval performance. The Mannville Group aquifer was selected due to its historic use as the water source for the oil and gas industry, favorable permeability, safe distance from drinking water sources, while close enough to surface, making it cost-effective for drilling.

## Methodology

**Understanding the Geological Background**

The Western Canadian Sedimentary Basin (WCSB) spans several provinces in Western Canada, including southern Saskatchewan. The WCSB offer significant thermal potential for energy storage, power and direct-use applications (Shokri et al., 2022). The deepest and thickest wedge of the basin lies along the Rocky Mountains as its western edge, gradually thinning eastward until it reaches the Canadian Shield, where it tapers to zero thickness at the surface. WCSB covers approximately two-thirds of southern Saskatchewan (Jessop & Vigrass, 1989; Majorowicz & Grasby, 2021).

A sub-sedimentary basin within the WCSB, recognized as the Williston Basin, underlies Saskatchewan and is composed of five primary transgressive-regressive sequences from the Cambrian to the Tertiary periods. These sequences reflect alternative depositional layers of sandstone, carbonate, and shale.

Hydro-stratigraphic studies of the Williston Basin have identified three main regional aquifers and aquitards, including the Lower Paleozoic, Mississippian, and Mesozoic units. Within the Mesozoic Group, the Mannville Group or Aquifer exhibits the highest permeability. It consists of interbedded sandstones and shales and reaches a maximum thickness of approximately 150 m. It is overlain by the very thick Colorado-Lea Park aquitard.

Since 1990, the Mannville Group has been utilized both as a source of water and as a disposal formation for recovered fluids for oil and gas operations. Over the last decade, however, the volume of fluids injected into the Mannville Group has exceeded the volume of water produced from it (Jellicoe, 2020; Ufondu, 2017; Woroniuk, 2023).

**Construction of Geological Model**

A geomodel was built in Computer Modelling Group (CMG) software for the Mannville Group, underlying the City of Regina area, covering a 25 km × 15 km region. The reservoir was discretized into a gird with 160 × 160 m cell size in X and Y direction and divided into 10 layers in the Z direction. To account for confining shale layers, the reservoir model was extended vertically using available data on seal layers thicknesses.

Geological data for the Mannville Group were obtained from the logs of four existing wells in the target area. Table 1 provides detailed well information, including coordinates, measured depth (MD), true vertical depth (TVD), kelly bushing elevation, and the depth of formation tops. To identify formation tops and horizons, gamma ray logs were interpreted for all wells and are shown in the well section (Figure 1).

**Table 1—list of studied wells with the logistic and target formation information at the City of Regina area (Original Authors' work)**

| Well Name | Well ID | KB (m) | TVD (m) | MD (m) | Mannville Group | Jurassic Group |
|---|---|---|---|---|---|---|
| UofR | 131/03-08-017-19W2/02 | 580.9 | 2225.6 | 2225.6 | 703.73 | 788.14 |
| Co-Op | 191/04-04-018-19W2 | 594.5 | 2198.5 | 2376 | 754.80 | 852.01 |
| TRANSGAS-9-22 | 101/05-23-18-19W2 | 612.8 | 2232.0 | 2232.0 | 896.72 | 800.67 |
| TRANSGAS-5-23 | 101/09-22-018-19W2 | 610.4 | 2195.0 | 2195.0 | 896.72 | 769.28 |

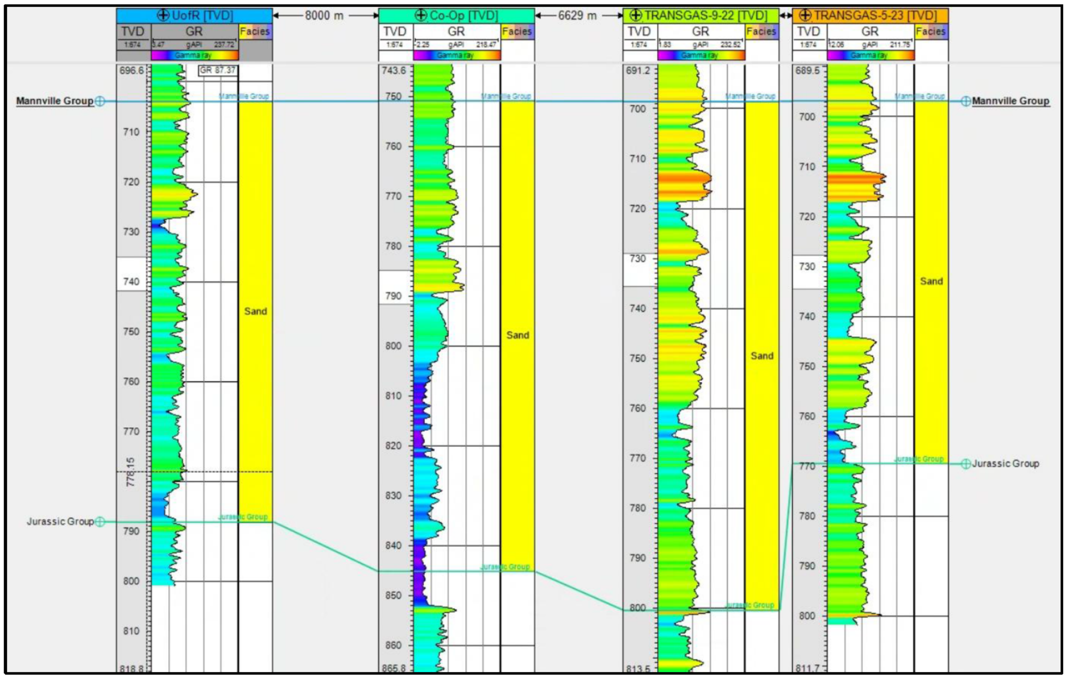


**Figure 1—Gamma ray log of the studied wells and the identified horizon of the Mannville Group and Jurassic Group (Original Authors' work)**

Identification of formation horizons from the available wells facilitated the development of structure maps. These maps, shown in Figure 2, correspond to the locations of the wells.

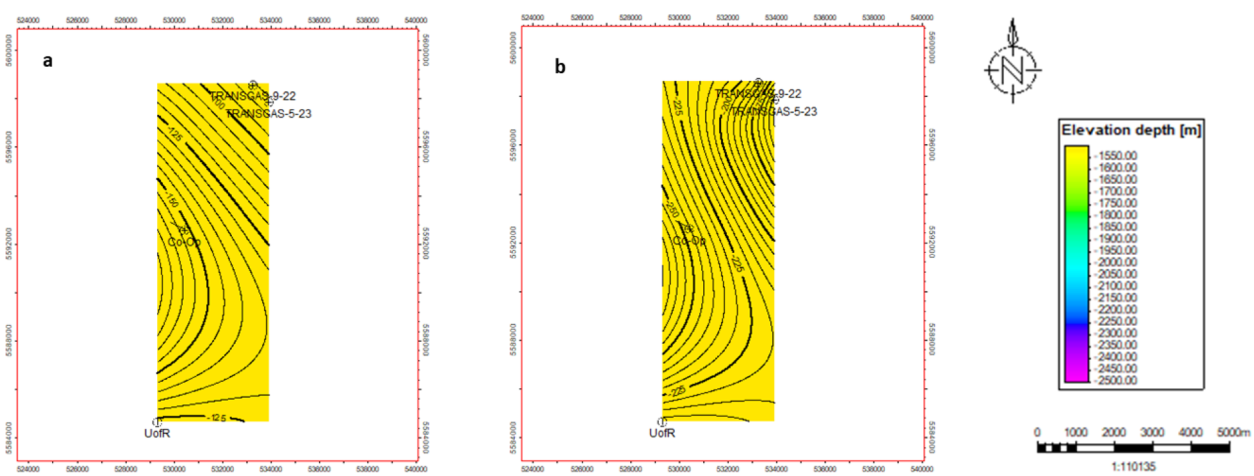


**Figure 2—Structure map of a) Mannville Group, b) Jurassic Group (Original Authors' work)**

Reservoir properties, including the physical and thermal characteristics of reservoir fluids and rocks, were collected from studies conducted on the Mannville Group in southeastern Saskatchewan (Jellicoe, 2020;

Ufondu, 2017; Woroniuk, 2023). Additional parameters were interpreted from core analyses and well log curves, such as density-porosity relationships in sandstone formations, used for porosity estimation.

Permeability was calculated as a function of porosity and temperature distribution using a formula provided by Vigrass et al. (2007), which was applied to construct the reservoir model shown in Figure 3. Permeability was calculated as a function of porosity and temperature distribution using a formula provided by Vigrass et al. (2007), which was applied to build the reservoir model shown in Figure 3.

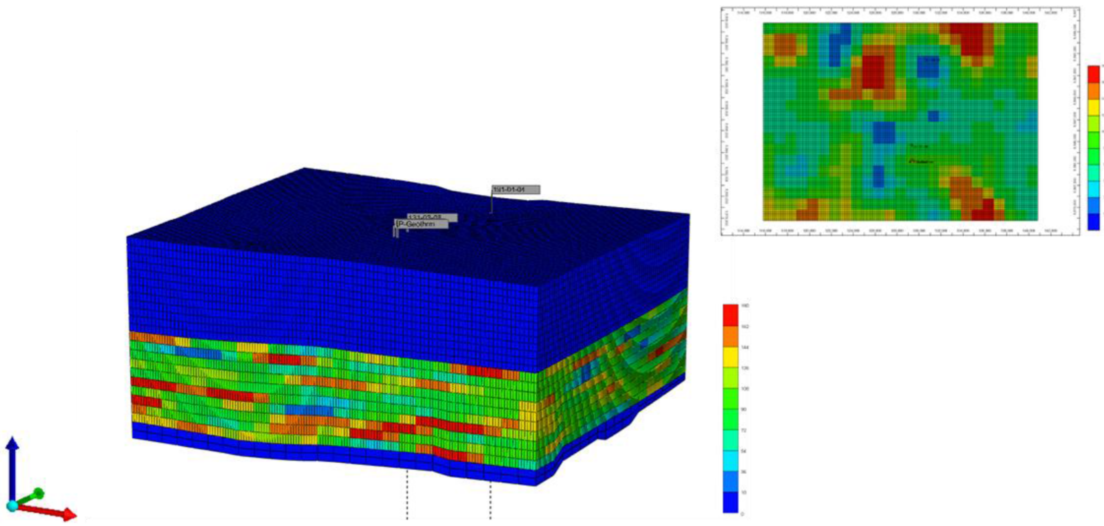

**Figure 3—3D model of the Mannville Group and a display of the porosity distribution from a layer of geomodel (Original Authors' work)**

Permeability = (697.74 * porosity) - 28.964
Temperature = (0.027 * depth) + 26

## Solar Thermal Integration for Geothermal Reservoir Modeling

Shafieian et al. (2019) conducted a study on the performance of thermal energy storage using an evacuated tube solar pilot set-up. The study showed that the thermal efficiency of the solar heating system depends on the number of glass tubes within each solar pipe rack of the heat collecting system, with an optimal value and configuration identified. A set-up consisting of 25 evacuated solar tubes was used, which achieved a water temperature of approximately 60°C under a solar radiation intensity of 376.7 W/m$^2$.

Solar energy data, including monthly average irradiation in kWh/m$^2$ and monthly average sunshine hours, for the City of Regina, were obtained from a publicly accessible source (https://online.tsol.de/en/) and are presented in Figure 4. Only months with peak radiation values were considered for energy storage operations. It was assumed that in the month of May, with 265 sunshine hours and the maximum average irradiation of 205 kWh/m$^2$ for the year, the solar system would be capable of harvesting water at 60 °C. Solar energy availability in other months was estimated relative to the radiation conditions in May.

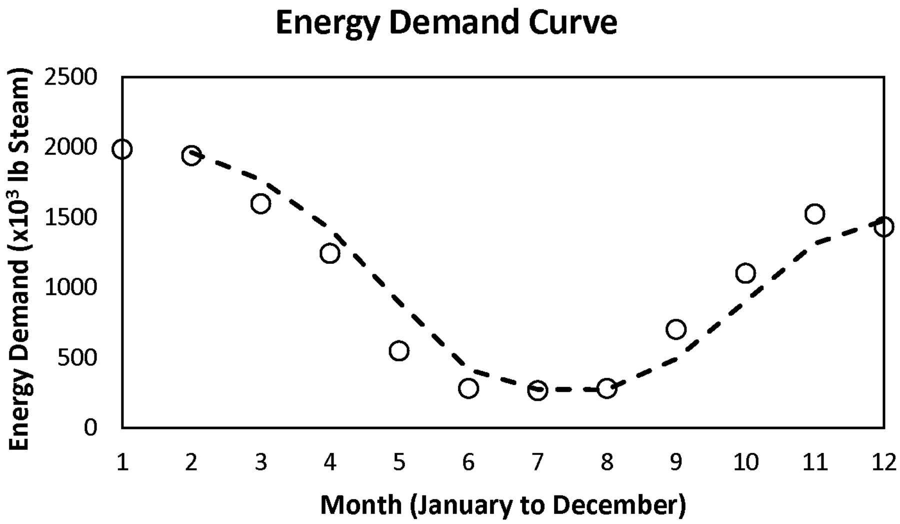


**Figure 4—Monthly energy demand from a facility located at the studied region (Original Authors' work)**

The energy demand pattern for producing from the reservoir was based on data provided by a facility located at the University of Regina campus, also shown in Figure 4. All relevant solar radiation values, as well as storage and extraction flow rates, are summarized in Table 2.

**Table 2—Solar irradiation information and the storage and production flow rate from the Mannville reservoir (Original Authors' work)**

| Month | KWh/m² | Monthly Average Sunshine (hr) | Injection (m³/day) | Production (m³/day) | Approximate Temperature Increase (°C) |
|---|---|---|---|---|---|
| January | - | - | - | 1500 | - |
| February | - | - | - | 1500 | - |
| March | 125.491 | 158 | 1000 | 1300 | 37.76 |
| April | 164.886 | 239 | 1000 | 1000 | 50.37 |
| May | 205.169 | 265 | 1000 | SHUTIN | 60 |
| June | 178.372 | 280 | 1000 | SHUTIN | 57.157 |
| July | 196.163 | 327 | 1000 | SHUTIN | 66.3 |
| August | 165.692 | 290 | 1000 | SHUTIN | 56 |
| September | 127.839 | 199 | 1000 | SHUTIN | 41.4 |
| October | - | - | - | 900 | - |
| November | - | - | - | 1200 | - |
| December | - | - | - | 1100 | - |

The wells would be spaced 400 m apart to optimize access to the hot plume generated during injection. Injection operations were assumed to begin in March 2026, utilizing temperatures harvested from the solar thermal system, described in Table 2. A constant flow rate of 1000 m³/day would be applied during months with higher solar radiation.

To allow sufficient time for the hot plume to form and expand within the reservoir, production would be postponed until January 2027, aligned with the anticipated energy demand. During certain months, production would be paused due to lower energy requirements. Both the storage and production processes would continue in operation until the end of 2040.

## Results and Discussion

The primary focus of this study is to investigate the impact of periodic hot water storage, sourced from a solar thermal system, on subsurface aquifer temperature. Figure 5 shows the temperature of produced water both at the bottomhole and at the surface. Over time, both temperatures increase: bottomhole temperature rises from 32.8 °C to 34 °C while surface temperature increases from 30.6 °C to 31.7 °C.

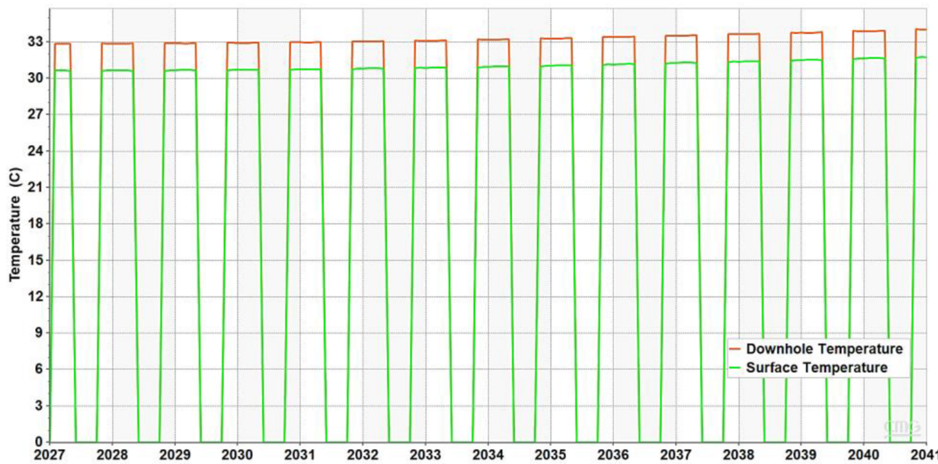


**Figure 5—Downhole and surface temperature variation affected by storing solar thermal energy (Original Authors' work)**

Two key observations are drawn from the results:

Although the storage process would not be continuous, since it was only for seven months of the year and occasionally overlap with production months, a considerable rise in reservoir temperature could be achieved.

The wells would be spaced approximately 400 m apart, constrained by grid size limitations. As shown in Figure 6, the production well will lie within the hot plume, though the highest temperature zone would not yet be reached in the production well.

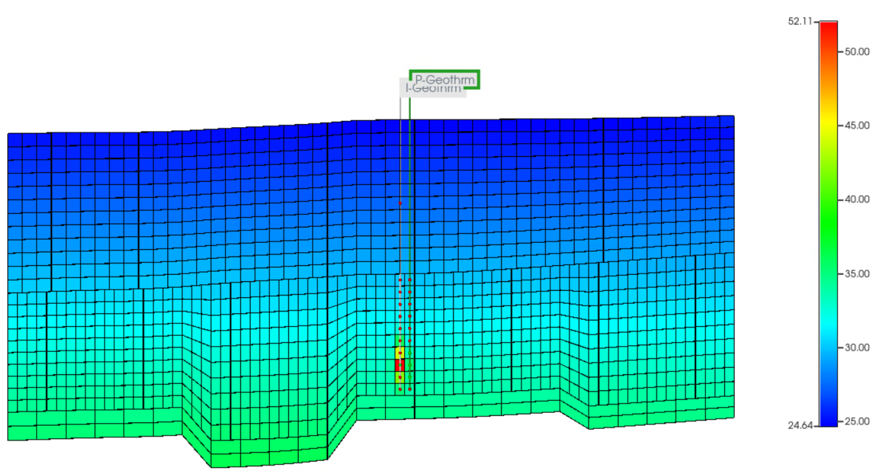


**Figure 6—Hot plume expansion into the reservoir with the well's placement respective to the plume (Original Authors' work)**

This suggests that placing the production well closer to the storage well could yield water with higher temperatures. Notably, the storage binary system does not necessarily follow the conventional geothermal doublet design, which typically requires a minimum 1 km distance between wells.

Figure 7 highlights the increase in harvested energy resulting from the reservoir temperature enhancement due to thermal storage. The accessible energy at the surface is projected to increase from 60 kJ/kg to 65 kJ/kg of produced water by the end of 2040. This increase indicates that, even with the same amount of produced water, the storage of surplus solar energy enables extraction from a reservoir with a higher initial energy and temperature compared to the beginning of the process.


**Figure 7—Variation of the reservoir fluid enthalpy at the bottomhole and on the surface in 14 years of operation (Original Authors' work)**

Figure 8 shows that the cumulative energy stored through the injection of surplus solar thermal energy during the operation period would be nearly twice the amount of energy produced from the reservoir. The modeled ATES system demonstrates that meeting the facility's energy demand would not deplete reservoir temperature. On the contrary, the temperature would increase over time as additional energy was continuously fed into the system.

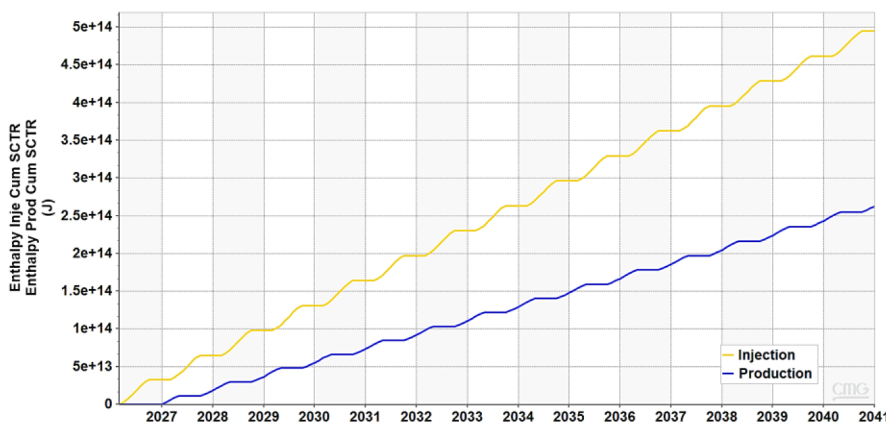


**Figure 8—Cumulative energy injected and produced from the ATES reservoir (Original Authors' work)**

## Conclusions and Remarks

A substantial portion of residential energy demand is attributed to domestic heating, which is predominantly met by burning fossil fuels. Renewable energy resources, such as solar, wind, biomass, and geothermal, offer a sustainable alternative, contributing both to energy security and $CO_2$ emission reduction. Among these, seasonal energy storage systems like aquifer thermal energy storage (ATES) present a promising solution, enabling integration of geothermal resources with surplus solar or industrial waste heat. This intermittent storage approach helps mitigate the mismatch between energy availability and demand.

This study developed a geomodel of a relatively shallow aquifer beneath the City of Regina, Saskatchewan, to evaluate the potential for storing solar energy harvested via an evacuated tube solar collector system. Results from the simulation showed that reservoir and production fluid temperatures would steadily increase over a 14-year operational period. Elevated reservoir temperatures could allow for greater energy extraction per unit of produced water.

The well spacing in the model was approximately 400 m, constrained by grid resolution. Temperature distribution plots indicated that the production well lies within the thermal plume, although the hottest region has not yet reached the wellbore. These findings suggest that reducing the inter-well distance could improve energy yield, especially since ATES systems do not adhere to the conventional 1 km separation required in geothermal doublet configurations.

By the end of 2040, the cumulative stored energy would be approximately twice that of the produced energy, indicating continuous reservoir charging and a net increase in subsurface temperature over time. This underscores the long-term viability and performance of solar-assisted ATES for residential hearing applications in cold-climate regions.

## Acknowledgments

We also acknowledge Schlumberger for supplying the Petrel software package, and Computer Modelling Group Ltd. (CMG) for providing the CMG software suite. Authors of this paper would like to extend their sincere appreciation to Saskatchewan Geological Survey and geoLOGIC Systems Ltd. for their persistent support in providing geological data.

## Nomenclature

| | |
|---|---|
| ATES | Aquifer thermal energy storage |
| BTES | Borehole Thermal Energy Storage |
| CMG | Computer Modelling Group |
| $CO_2$ | Carbon Dioxide |
| GHG | Greenhouse Gas |
| KB | Kelly Bushing Elevation |
| MD | Measured Depth |
| TES | Thermal Energy Storage |
| TVD | True Vertical Depth |

## Units

| | |
|---|---|
| Dimension | m |
| Flow Rate | $KWh/m^2$ |
| Permeability | mD |
| Temperature | °C |
| Solar Radiation Intensity | m3/day |

## References

Beernink, S., Hartog, N., Vardon, P. J., & Bloemendal, M. (2024). Heat losses in ATES systems: The impact of processes, storage geometry and temperature. *Geothermics*, 117. https://doi.org/10.1016/j.geothermics.2023.102889

Daniilidis, A., Mindel, J. E., De Oliveira Filho, F., & Guglielmetti, L. (2022). Techno-economic assessment and operational CO2 emissions of High-Temperature Aquifer Thermal Energy Storage (HT-ATES) using demand-driven and subsurface-constrained dimensioning. *Energy*, 249. https://doi.org/10.1016/j.energy.2022.123682

Daniilidis, A., Scholten, T., Hooghiem, J., De Persis, C., & Herber, R. (2017). Geochemical implications of production and storage control by coupling a direct-use geothermal system with heat networks. *Applied Energy*, **204**, 254–270. https://doi.org/10.1016/j.apenergy.2017.06.056

Jellicoe, K. (2020). QUANTIFYING PRODUCED AND INJECTED WATER VOLUMES IN SOUTHEASTERN SASKATCHEWAN. University of Saskatchewan.

Jessop, A. M., & Vigrass, L. W. (1989). Geothermal measurements in a deep well at Regina, Saskatchewan. In *J. Volcanol. Geotherm. Res* (Vol. **37**).

Majorowicz, J., & Grasby, S. E. (2021). Deep geothermal heating potential for the communities of the western Canadian sedimentary basin. *Energies*, **14**(3). https://doi.org/10.3390/en14030706

Mesquita, L., McClenahan, D., Thornton, J., Carriere, J., & Wong, B. (2017). Drake Landing solar community: 10 years of operation. ISES Solar World Congress 2017 - IEA SHC International Conference on Solar Heating and Cooling for Buildings and Industry 2017, Proceedings, 333–344. https://doi.org/10.18086/swc.2017.06.09

Movahedzadeh, Z., Shokri, A. R., Chalaturnyk, R., Nickel, E., & Sacuta, N. (2021). Measurement, monitoring, verification and modelling at the Aquistore CO2 storage site. *First Break*, **39**(2), 69–75.

Rangriz Shokri, A., Hau, K. P., Saar, M. O., White, D., Nickel, E., Siddiqi, G., & Chalaturnyk, R. (2021). Modeling CO2 Circulation Test for Sustainable Geothermal Power Generation at the Aquistore CO2 Storage Site, Saskatchewan, Canada. 2nd Geoscience & Engineering in Energy Transition Conference.

Shafieian, A., Khiadani, M., & Nosrati, A. (2019). Thermal performance of an evacuated tube heat pipe solar water heating system in cold season. *Applied Thermal Engineering*, **149**, 644–657. https://doi.org/10.1016/j.applthermaleng.2018.12.078

Shokri, A. R., Vaisblat, N., Wigston, A., Champollion, Y., Hickson, C., & Chalaturnyk, R. (2022). Improving the Competitiveness of Geothermal Energy Exploitation through Integrated CO2 Storage in the Western Canadian Sedimentary Basin. *Geothermal Rising*.

Sibbitt, B., Mcclenahan, D., Djebbar, R., Thornton, J., Wong, B., Carriere, J., & Kokko, J. (n.d.). MEASURED AND SIMULATED PERFORMANCE OF A HIGH SOLAR FRACTION DISTRICT HEATING SYSTEM WITH SEASONAL STORAGE.

Sibbitt, B., McClenahan, D., Djebbar, R., Thornton, J., Wong, B., Carriere, J., & Kokko, J. (2012). The performance of a high solar fraction seasonal storage district heating system - Five years of operation. *Energy Procedia*, **30**, 856–865. https://doi.org/10.1016/j.egypro.2012.11.097

Siuta-Olcha, A., Cholewa, T., & Dopieralska-Howoruszko, K. (2020). Experimental studies of thermal performance of an evacuated tube heat pipe solar collector in Polish climatic conditions. https://doi.org/10.1007/s11356-020-07920-3/Published

Talman, S., Shokri, A. R., Chalaturnyk, R., & Nickel, E. (2020). Salt precipitation at an active CO2 injection site. *Gas Injection into Geological Formations and Related Topics*, 183–199.

Ufondu, L. S. (2017). ASSESSMENT OF THE POTENTIAL HEAT STORED IN THE DEEP AQUIFERS OF THE WILLISTON BASIN FOR GEOTHERMAL ENERGY PRODUCTION.

Vigrass, L., Jessop, A., & Brunskill, B. (2007). Regina geothermal project. *Summary of Investigations*, **1**, 2004–2007.

Vrijlandt, M., Dinkelman, D., Otero Rodriguez, G. J., & Koenen, M. (2023). ThermoGIS-HTO National high temperature aquifer thermal energy storage potential tool ThermoGIS-HTO.

Woroniuk, B. W. (2023). HYDROGEOLOGY OF THE INTERMEDIATE ZONE BETWEEN DEEPER ANTHROPOGENIC ACTIVITIES AND SHALLOW AQUIFERS IN SASKATCHEWAN. University of Saskatchewan.

Zwamborn, M., Beernink, S., Kleinlugtenbelt, R., Oerlemans, P., Pothof, I., Star, K., Dinkelman, D., Van Bergen, F., & Bloemendal, M. (2022). HT-ATES systems in district heating networks, a Dutch benchmark study. In *European Geothermal Congress*. www.europeangeothermalcongress.eu